STUDY- Factors of significance in storm formation

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Submitted by

Karthik Cherukuru (AP23110010658)

Krishna Kireeti Vemuri (AP23110010734)

Satwik Paxton Dasari (AP23110011268)

Gollapalli Sai Manish (AP23110010940)

Jampani Siva Krishna (AP23110010838)

Under the Guidance of

Dr Suraj Kumar Bhagat

SRM University–AP

Neerukonda, Mangalagiri, Guntur

Andhra Pradesh – 522 240

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1. INTRODUCTION

The development of storms and unprecedented precipitation is one of the most influential weather processes in the coastal and tropical areas. Over the past few decades, growing variability of climate, explosive urbanization, and dynamics in atmospheric processes have increased the intensity, frequency, and severity of intense rain events. Such occurrences have a great danger on infrastructures, agriculture, disaster management structures and safety. As a result, the capability of scientifically studying and forecasting storm-related rainfall through data-driven approaches has gained a significant interdisciplinary research subject on the boundary between meteorology and applied data science.

The following is an applied analysis study that is intended to determine the meteorological factors of significance in storm formation through precipitation intensity as a proxy variable of storm events. The discussion is based on practical atmospheric data obtained at NASA POWER database, a cyclone-prone area is considered to ensure that the analysis is relevant to the context. The project aims to bridge the gap between theoretical meteorological concepts and predictive models based on computation with the use of statistical exploration, visualization techniques, feature selection methodologies, and machine learning models.

This study is driven in part by the increased relevance of evidence-based climate risk assessment and early warning systems. Conventional meteorological forecasting is highly dependent on physical models and domain knowledge, but the rapid increase in the volume of available environmental data has allowed the data science method to be used to identify latent patterns and predictive indicators. This project is thus designed to follow a systematic analysis pipeline; i.e., data acquisition and preprocessing, exploratory data analysis (EDA), high-fidelity feature selection by filter plus wrapper approach, and finally, comparative machine learning model evaluation.

Moreover, the paper focuses on interpretability and scientific validation. In contrast to the act of prediction which is a purely computational task, the analysis links identified statistical relationships between observed indicators of atmospheric moisture and the impact of pressure variations with known meteorological theory about storm actions and rainfall generation. This compound viewpoint makes it so that the conclusions that are obtained are both physically significant and statistically sound.

After all, the goal of this project is two-fold:

The first and the second questions to be answered are; first, to identify the atmospheric variables that have the strongest correlation with heavy rainfall events, and second, to determine the predictive persistence of various machine learning models under realistic class imbalance. In such a manner, the project helps to gain a more comprehensive insight into the patterns of precipitation occurring during storms and proves the practicability of the applied methods of data science in the field of environmental analysis.

2. PROBLEM DEFINITION

Extreme events of precipitation and storm-related rainfall have now become key issues of environmental concern, especially in the coastal and tropical areas of the world where atmospheric turbulence, high moisture content, and active pressure systems come together to interact with each other often. Although big data sets of meteorological data are available, correctly determining the dominant atmospheric drivers of storm formation is an intricate analytical task. Conventional forecasting methods are very dependent on the use of numerical weather prediction models and domain heuristics, but they are not always able to make the most of latent statistical relationships that exist within long-term observational data.

The fundamental issue that this project aims to solve is to scientifically identify and numerically assess meteorological variables that have a strong effect on storms by considering heavy precipitation events as a working proxy measure of storms. In particular, the research aims at establishing the interaction of atmospheric moisture cues (i.e. specific humidity, relative humidity, dew point), thermodynamic factors (temperature, wet-bulb temperature), dynamic ones (wind speed, wind direction) and synoptic-scale ones (varying surface pressure) to generate high-intensity rain episodes.

One of the issues in this context of problems is that there is an imbalance between classes and the rarity of events. Extreme rainfall events are only a minor fraction of all observations in real-world weather datasets, and thus predictive modeling is challenging. Models that are trained with this kind of data are prone to over-fitting to the prevalent conditions that are not storms and thus attaining a misleadingly high accuracy but not identifying any critical storm events. The reduction of this imbalance by means of proper threshold engineering, evaluation measures, and model choice will thus be part and parcel of the analytical problem.

The redundancy of features and the contextuality of time is another aspect of the issue. In meteorological datasets, it is often the case that much of the predictive value of some variables is indirect, not causal. It is crucial to separate the variables that are genuinely predictive of storm formation and those that are only reflectively or seasonally correlated so as to develop predictive frameworks that are interpretable and reliable.

After all, the issue is more than just prediction accuracy. The project will focus on establishing a methodologically sound, interpretable and data based knowledge of how storms are formed, which will allow making insights into the formation mechanisms of these storms in accordance with the well known principles of atmospheric science. Through incorporation of exploratory visualization, statistical feature identification, and comparative machine learning assessment, the research is aimed at converting the raw environmental data into actionable knowledge on the processes of heavy rainfall events.

3. PROBLEM STATEMENT

Extreme rainfall events and storm-related precipitation are becoming more frequent and are causing severe environmental, infrastructural, and socio-economic dilemmas especially in areas that are susceptible to the tide. Even with the current developments in the field of meteorological observation and access to high-resolution datasets of the atmosphere, the most critical missing link in the field is the possibility of identifying the most contributing meteorological factors that cause a storm formation through data-driven methods of analysis. This restriction impacts on the formulation of viable predictive models that can be used to advance early warning mechanisms, catastrophe preparedness, and measures to mitigate climate risks.

The main issue that is discussed in this work is insufficient, quantitatively justified knowledge of the interrelationship between various factors in the atmosphere to result in heavy precipitation events that are often employed as operational measures of storm activity. Although meteorological theory indicates that atmospheric moisture contents, changes in surface pressure, temperature difference, and wind dynamics are factors of some significance in the development of storms, the relative significance of these factors, and their predictive power, may differ greatly from one region to another, and over time. As a result, an empirical, data-based study, which will assess these variables in a framework of an organized analytical framework, is required.

The issue of modelling rare but high impact events is another significant aspect of the problem. Extreme precipitation events in long term weather observations normally form a small portion of the total observations leading to skewed data distributions. This skew makes machine learning model training problematic, since more traditional evaluation measures such as accuracy might not be indicative of a model's actual capacity to predict storm events. Thus, not only the selection of the variables but also the methodological rigor that is needed to provide meaningful model evaluation and comparison must be discussed in the study.

Also, the meteorological data is often characterized by variables that are temporally dependent and seasonally modulated, and which can be statistically correlated, but does not necessarily have causal importance in the storm development. Separating the real atmospheric drivers and the contextual, or redundant, predictors is necessary so as to construct interpretable and scientifically consistent predictive models.

In this respect, the problem statement that will inform this project is to come up with an analytical pipeline that will realize the most critical meteorological predictors of heavy rainfall caused by storms and determine the efficiency of various machine learning methods in effectively classifying such occurrences. In such a way, the research should assist in the effort to enhance data-centered information on storm processes and aid the overall objective of enhancing the capacity to assess environmental risk and forecasting.

4. OBJECTIVES

- To interpret the meteorological information over a long period of time taken via the NASA POWER database with a view to learn atmospheric patterns as related to rain that is heavy.
- To operationalize storm occurrence based on a scientifically determined precipitation level.
- To make sure that extreme precipitation events are not arbitrarily defined.So as to carry out extensive exploratory data analysis (EDA) and visualization, such as distribution analysis, storm vs non-storm comparisons, seasonal trend determination, and time-series pattern analysis.
- To determine the strongest atmospheric variables that cause a storm to occur through studying the signs of moisture (specific humidity, relative humidity, dew point), thermodynamic variables (temperature, wet-bulb temperature), changes in pressure, and wind properties.
- To use and compare several feature selection methods, such as filter methods (correlation analysis, mutual information, variance threshold), and wrapper methods (Recursive Feature Elimination and forward feature selection), to guarantee the robust and consistent identification of predictors.
- To develop and analyze several machine learning classification models including Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting to predict a storm.
- To evaluate the performance of the model with the help of proper evaluation measures such as Precision, Recall, F1-Score, and ROC-AUC, especially with realistic class imbalance scenarios.
- To be able to make interpretations of the analytical results as per the recognized meteorological principles so that predictive understanding will in some way be statistically significant and scientifically meaningful.
- To eventually identify the most useful prediction model and the most influential atmospheric processes in storm-related heavy precipitation incidents.

5. METHODOLOGY

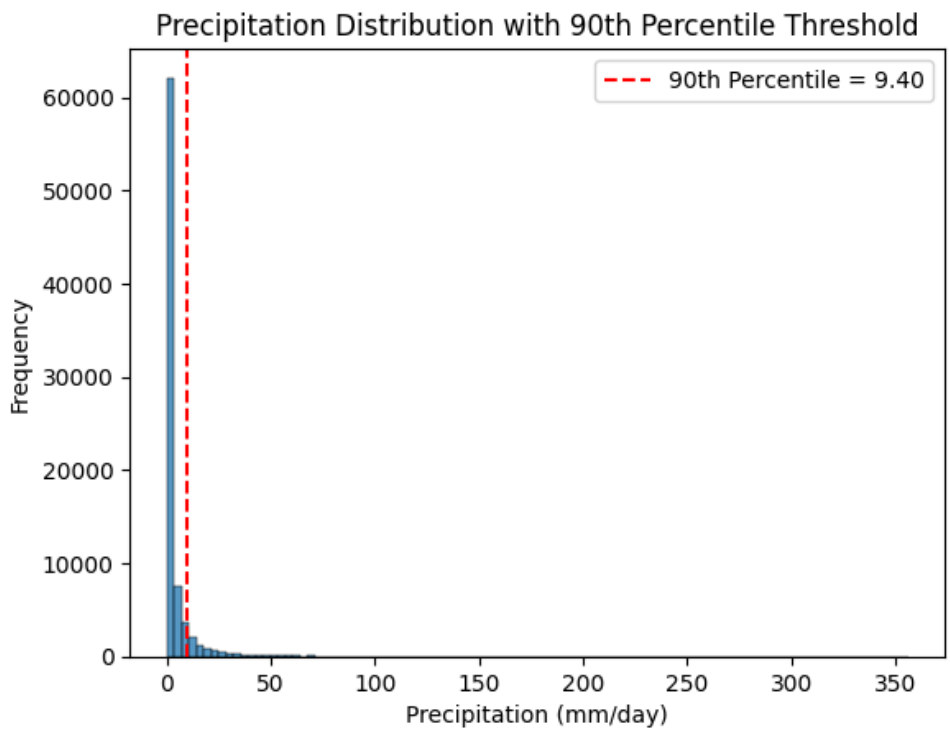
5.1 Data Collection and Construction of the Data.

The research commenced by obtaining hourly weather information through the NASA POWER database, in one of the coastal cyclone prone areas in order to guarantee relevant studies to the environment. The dataset that was downloaded included the atmospheric variables of **temperature** (T2M), **dew point** (T2MDEW), **wet-bulb temperature** (T2MWET), **relative humidity** (RH2M), **specific humidity** (QV2M), **surface pressure** (PS), **wind speed** (WS10M), **wind direction** (WD10M), and **corrected precipitation** (PRECTOTCORR). Initially the raw dataset contained metadata rows and descriptive headers which were stripped off programmatically during the loading using row- skipping methods. Following the process of consolidation, core features were reduced to 13 and the final working dataset was 80,376 observations, which would underpin the next portion of analysis.

5.2 Pre-processing and Cleaning of Data.

A preprocessing pipeline was made systematically in order to maintain reliability in the dataset. The sentinel missing values (-999) were substituted by NaN, and then validation checks were done, and no re-remaining missing values and duplicate records were found. Individual temporal attributes (year, month, day, hour) were combined to form a single datetime index, which allows a chronological sort and allows analysis of time-series. The names of columns were standardized to enhance interpretability to enable consistency in the stages of analyses.

```
Shape of dataset: (80376, 13)

✅ Columns renamed!

Missing values per column:
year             0
month            0
day              0
hour             0
temp             0
dew_point        0
wet_bulb         0
rel_humidity     0
spec_humidity    0
pressure         0
wind_speed       0
wind_dir         0
precipitation    0
dtype: int64

Number of duplicate rows: 0

✅ Datetime column created!

Max Precipitation Value: 355.96
Chosen Threshold (90th percentile): 9.4
```



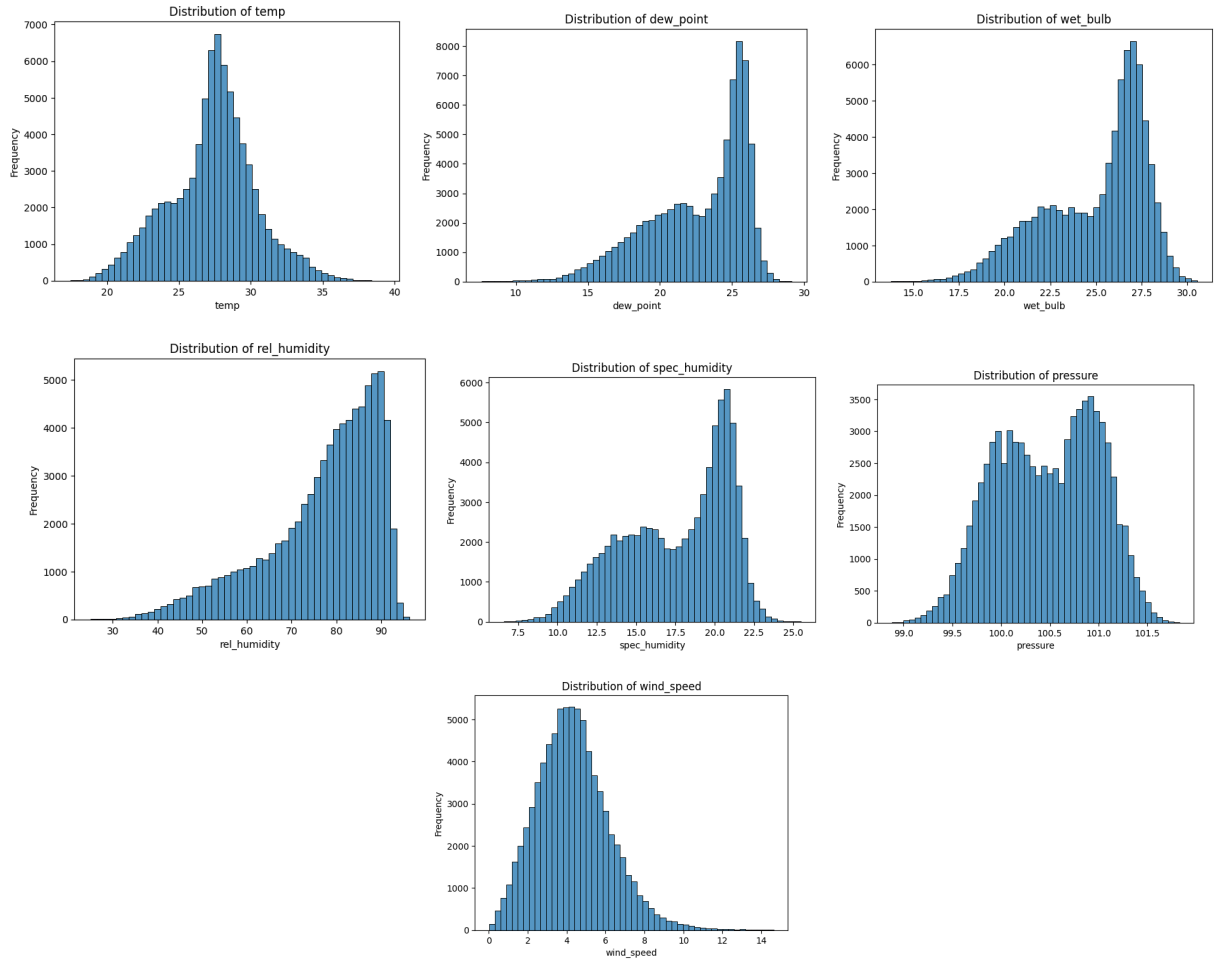
5.3 Target Engineering and Class Definition.

A data driven classification strategy was employed as opposed to arbitrarily defining the rainfall threshold. The precipitation distribution 90 th percentile (9.4 mm) was set as the determinant of storm events to ensure that only the maximum precipitation rates were categorized as storm events. Such a strategy created an effective simulation of a realistic class imbalance (approximately 90 percent non-storm and 10 percent storm) which is similar to the real environmental conditions and can be used in predictive modeling.

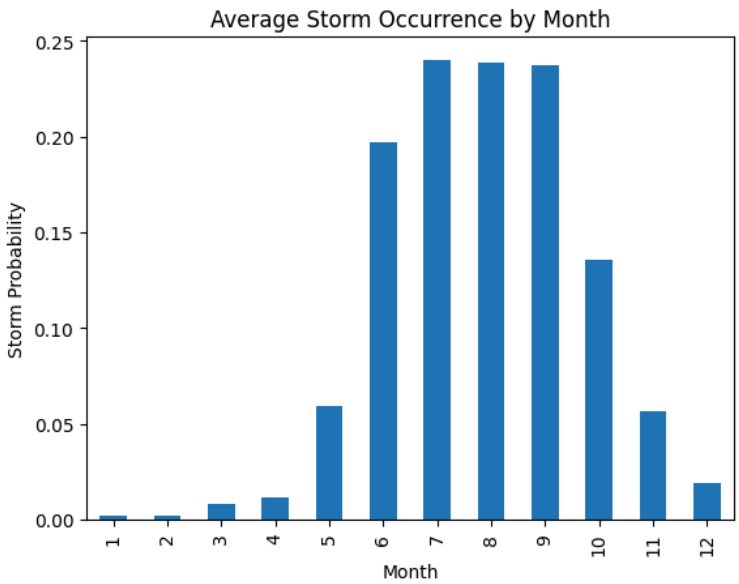
5.4 Exploratory Data Analysis and Visualization.

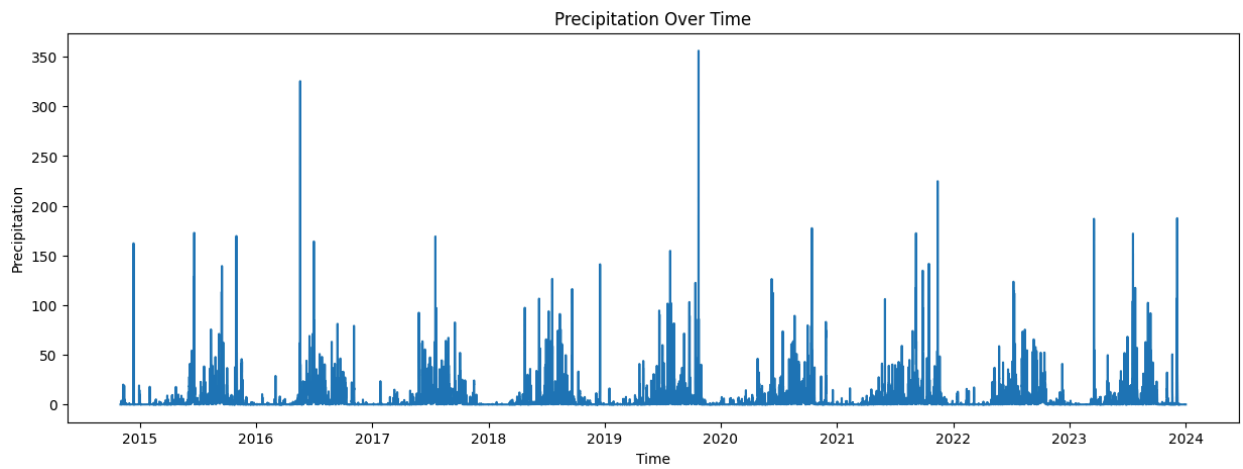
A lot of visualization was done to reveal the patterns in the atmosphere. The skewness of precipitation and humidity variables was seen in the distribution plots and the storm vs non-storm boxplots showed that, in case of a storm, there was more moisture in the atmosphere and lower pressure. Aggregation monthly revealed seasonal peaks as per

monsoon and cyclone activity. Time-series plots also showed there were extreme rainfall spikes of over 350 mm which confirmed that there were high impact weather events.



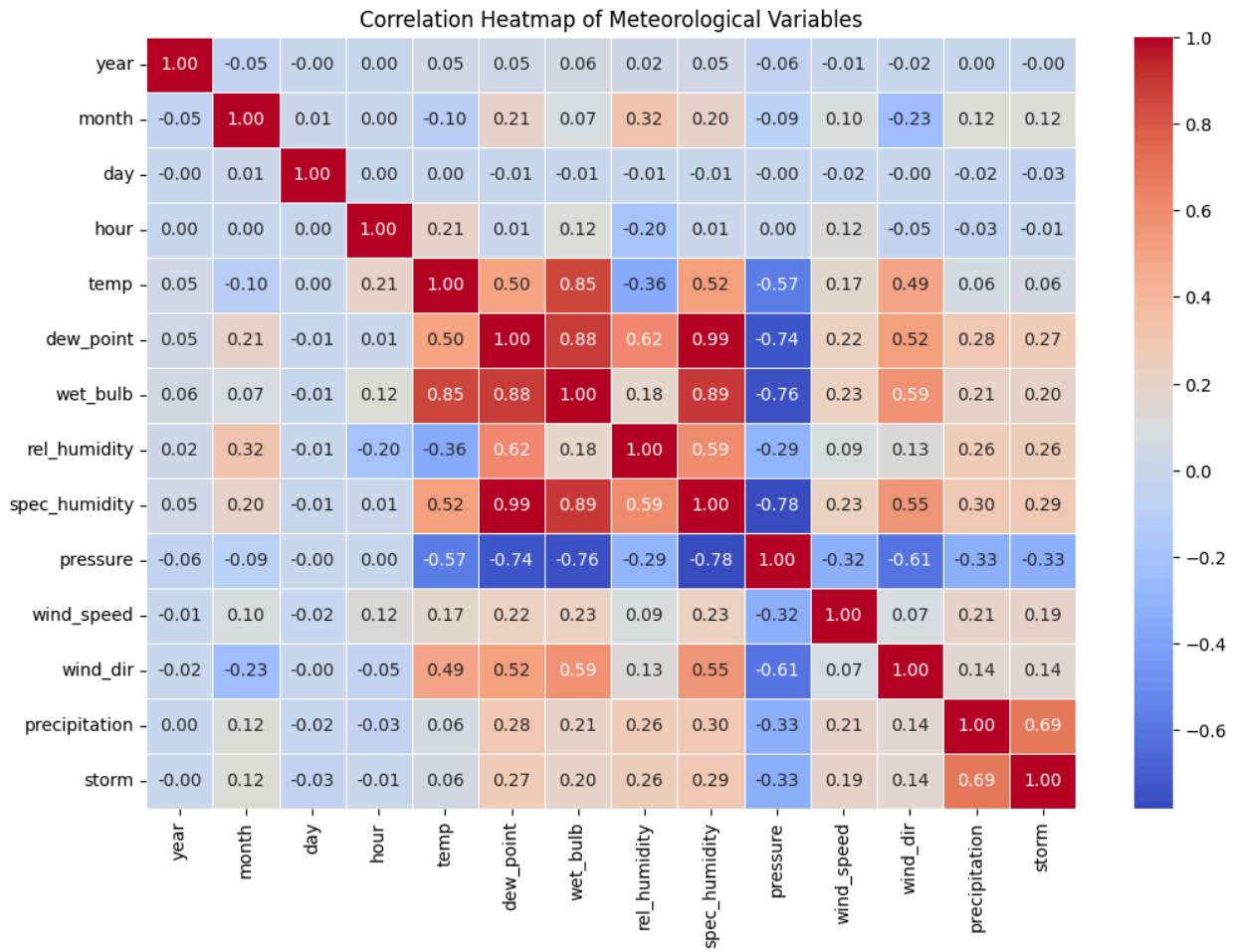
(DISTRIBUTIONS OF ALL FEATURES)





5.5 Selection of features and Analytical Modeling.

Various feature selection techniques were used. Correlation analysis and mutual information of storm occurrence and moisture indicators proved strong positive relationships, and also, a significant negative relationship with surface pressure. Wrapper techniques like Recursive Feature Elimination and forward selection verified the significance of having such variables as specific humidity, dew point, relative humidity, pressure, wet-bulb temperature, and wind speed.


Correlation Heatmap of Meteorological Variables
year 1.00 -0.05 -0.00 0.00 0.05 0.05 0.06 0.02 0.05 -0.06 -0.01 -0.02 0.00 -0.00
month -0.05 1.00 0.01 0.00 -0.10 0.21 0.07 0.32 0.20 -0.09 0.10 -0.23 0.12 0.12
day -0.00 0.01 1.00 0.00 0.00 -0.01 -0.01 -0.01 -0.01 -0.00 -0.02 -0.00 -0.02 -0.03
hour 0.00 0.00 0.00 1.00 0.21 0.01 0.12 -0.20 0.01 0.00 0.12 -0.05 -0.03 -0.01
temp 0.05 -0.10 0.00 0.21 1.00 0.50 0.85 -0.36 0.52 -0.57 0.17 0.49 0.06 0.06
dew_point 0.05 0.21 -0.01 0.01 0.50 1.00 0.88 0.62 0.99 -0.74 0.22 0.52 0.28 0.27
wet_bulb 0.06 0.07 -0.01 0.12 0.85 0.88 1.00 0.18 0.89 -0.76 0.23 0.59 0.21 0.20
rel_humidity 0.02 0.32 -0.01 -0.20 -0.36 0.62 0.18 1.00 0.59 -0.29 0.09 0.13 0.26 0.26
spec_humidity 0.05 0.20 -0.01 0.01 0.52 0.99 0.89 0.59 1.00 -0.78 0.23 0.55 0.30 0.29
pressure -0.06 -0.09 -0.00 0.00 -0.57 -0.74 -0.76 -0.29 -0.78 1.00 -0.32 -0.61 -0.33 -0.33
wind_speed -0.01 0.10 -0.02 0.12 0.17 0.22 0.23 0.09 0.23 -0.32 1.00 0.07 0.21 0.19
wind_dir -0.02 -0.23 -0.00 -0.05 0.49 0.52 0.59 0.13 0.55 -0.61 0.07 1.00 0.14 0.14
precipitation 0.00 0.12 -0.02 -0.03 0.06 0.28 0.21 0.26 0.30 -0.33 0.21 0.14 1.00 0.69
storm -0.00 0.12 -0.03 -0.01 0.06 0.27 0.20 0.26 0.29 -0.33 0.19 0.14 0.69 1.00
year month day hour temp dew_point wet_bulb rel_humidity spec_humidity pressure wind_speed wind_dir precipitation storm
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6

	Feature	Selected	Ranking
7	rel_humidity	True	1
6	wet_bulb	True	1
5	dew_point	True	1
4	temp	True	1
9	pressure	True	1
8	spec_humidity	True	1
10	wind_speed	False	2
1	month	False	3
0	year	False	4
3	hour	False	5
2	day	False	6
11	wind_dir	False	7

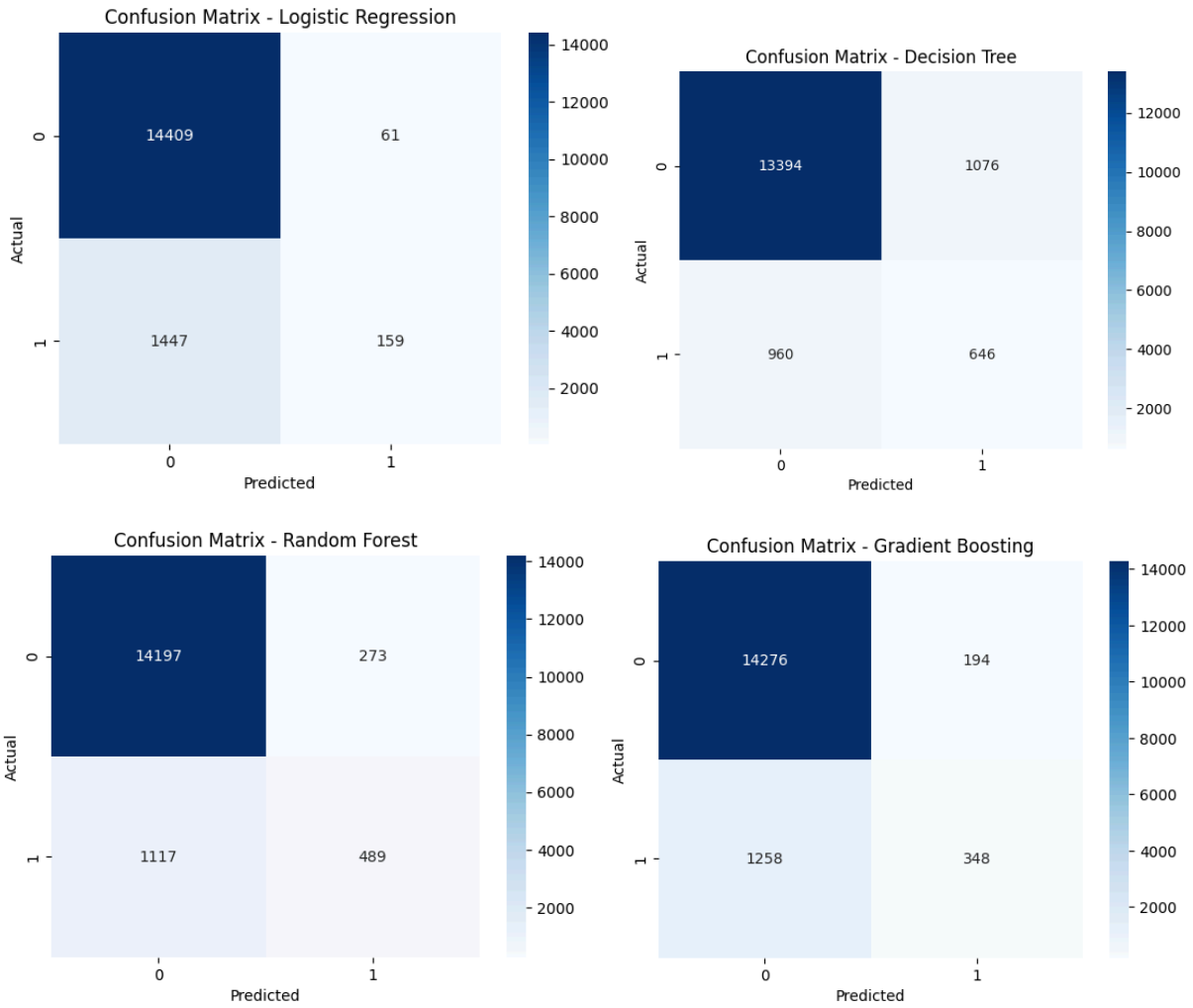
```
Correlation with Storm:

storm            1.000000
precipitation    0.688628
spec_humidity    0.287152
dew_point        0.272000
rel_humidity     0.257154
wet_bulb         0.197931
wind_speed       0.191541
wind_dir         0.138246
month            0.121325
temp             0.058678
year            -0.002647
hour            -0.014141
day             -0.032347
pressure        -0.327107
Name: storm, dtype: float64
```

	Feature	MI Score
8	spec_humidity	0.061413
5	dew_point	0.060975
1	month	0.059197
9	pressure	0.057307
7	rel_humidity	0.053439
6	wet_bulb	0.045247
4	temp	0.036630
10	wind_speed	0.019831
11	wind_dir	0.015287
0	year	0.002735
2	day	0.002229
3	hour	0.000262

```
Features after Variance Threshold:

Index(['year', 'month', 'day', 'hour', 'temp', 'dew_point', 'wet_bulb',
       'rel_humidity', 'spec_humidity', 'pressure', 'wind_speed', 'wind_dir'],
      dtype='object')
```

```
Forward Selected Features:

Index(['month', 'day', 'rel_humidity', 'pressure', 'wind_speed', 'wind_dir'], dtype='object')
```

Lastly, stratified train-test split was applied to maintain the distribution of classes and four classification models including logistic regression, decision tree, random forest, and gradient boosting were trained and assessed with The four measures of Precision, Recall, F1-Score, and ROC-AUC. The pipeline used was robust in identifying storm drivers as well as being scientifically interpretable.



6. RESULTS AND ANALYSIS

6.1 Characteristics and Distribution of Dataset and Events.

The ultimate analysis data was composed of 80,376 hourly meteorological measurements, spanning numerous years of one of the coastal areas prone to cyclones. The 90th percentile precipitation threshold (which is also known as 9.4 mm) was used as a data-driven storm classification strategy, with the outcome being about 10% storms and 90% non-storms. This distribution is realistic in that it represents the scarcity of extreme precipitation in the real atmosphere, and it provided a controlled imbalanced distribution of classes that affected the strategy of model evaluation.

Exploratory visualization indicated that precipitation had a high positive skewness and most values were concentrated around zero with a few extreme rainfall spikes with the highest amount of precipitation being 350mm and above, which is a result of extreme cyclonic-enhanced or monsoon-enhanced weather systems. Seasonal aggregation also showed probability of the occurrence of a storm in particular months in line with regional monsoon cycles which validated the environmental aspects in the data collected.

6.2 Feature Relationship Insights.

The statistical analysis was very convincing on the overriding atmospheric factors that cause storms. Correlation analysis revealed that specific humidity, dew point and relative humidity had the most significant positive correlation with storm occurrence, demonstrating the importance of moisture availability in the atmosphere during precipitation. On the other hand, surface pressure was found to have a significant negative relationship, which supports the meteorological theory of storm systems that tend to be characterized by low-pressure atmospheric conditions and a rise of vertical air.

Mutual Information analysis also found non-linear dependencies, which confirmed the fact that moisture indicators had a consistent effect even in the case of relaxed linear assumptions. These findings were confirmed by wrapper-based feature selection algorithms like Recursive Feature Elimination (RFE) which repeatedly selected variables of humidity and pressure as the central predictors. Other secondary predictors like wet-bulb temperature and wind speed were moderate in importance indicating their aiding role in the amplification of atmospheric instability and convergence of moisture.

The temporal variables such as year, day and hour exhibited a weak predictive value showing that storm formation is more influenced by synoptic atmospheric variables than isolated temporal variations.

6.3 Performance Comparison of Model Performance.

Gradient Boosting, Decision Tree, Logistic Regression and Random Forest were four classification models and were trained based on a train-test split with stratification to maintain the proportions of classes. Although the overall accuracy of all the models was approximately equal (90), as a metric it was not adequate because of the imbalanced nature of the storm occurrence.

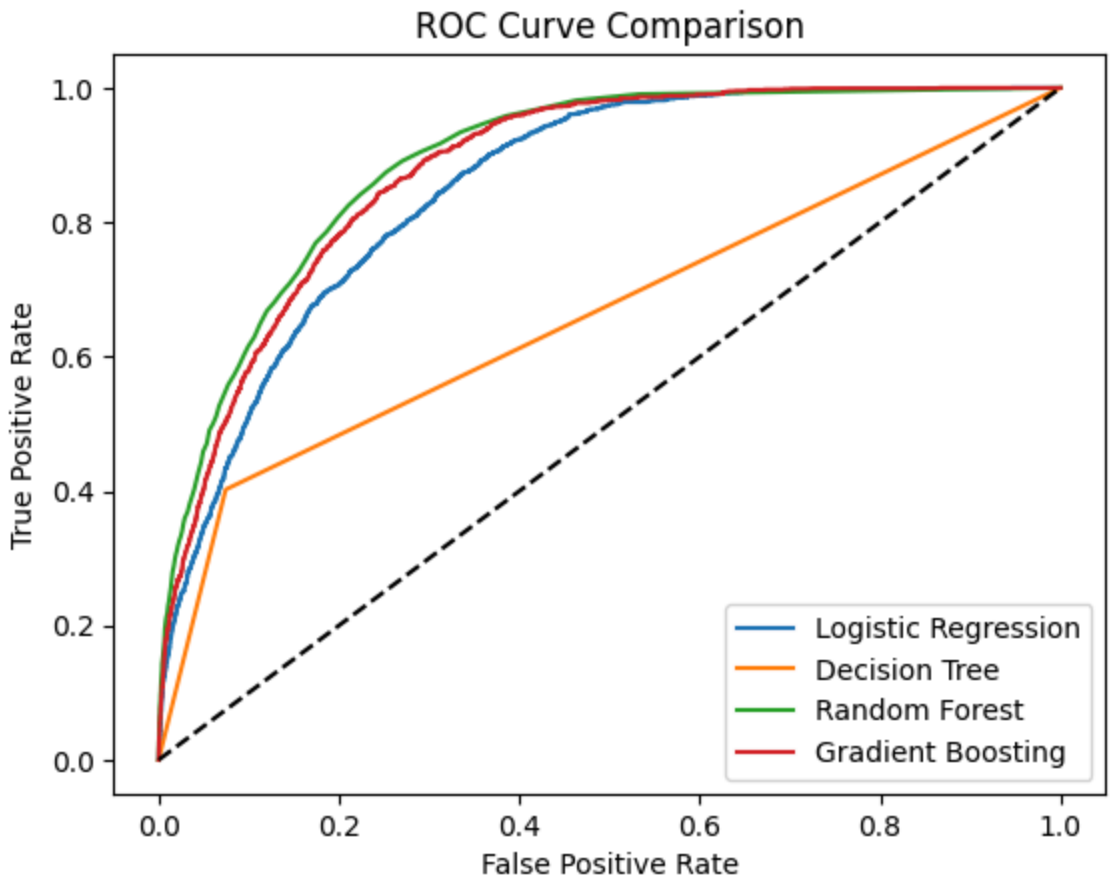
A less coarse analysis based on Precision, Recall, F1-Score, and ROC-AUC showed that there were strong variations in storm detection capacity:
Random Forest had the highest overall performance, the highest F1-Score (~0.41) and ROC-AUC (~0.89), which means that it was better at balancing false positives and missed storm events.

Decision Tree was the most sensitive to storm occurrences with Recall (~0.40) which implies that it is more sensitive to storms but at the expense of false alarms and overall reliability.

Gradient Boosting and the Logistic Regression had stronger precision and much lower recall which implies being conservative in their predictions and not predicting many real storm incidences.

These findings underscore the need to choose evaluation measures in accordance with real-world goals where it is more important to identify infrequent yet high-impact storms events than the overall accuracy.

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
Random Forest	0.913536	0.641732	0.304483	0.413007	0.89071
Decision Tree	0.873352	0.375145	0.402242	0.388221	0.66394
Gradient Boosting	0.909679	0.642066	0.216687	0.324022	0.879801
Logistic Regression	0.906196	0.722727	0.099004	0.174151	0.853067



6.4 Analytical Interpretation and Practical Implications.

The combination of statistical, visual and model-based analysis allows giving considerable grounds to believe that the content of moisture and pressure variation in the atmosphere are the most influential factors of heavy rainfalls occurrence in the considered area. Secondary factors, which enhance storm intensity, are wind dynamics and assortment of thermodynamic indicators like wet-bulb temperature.

Practically, the excellent performance of the Random Forest model implies that the ensemble learning strategies are especially useful in the meteorological interactions that are complex with little linearity. The results reveal that the application of data science methods may be used to augment conventional meteorological knowledge using quantitative metrics to confirm the formation mechanisms of storms and to provide predictive conclusions that can be used in the context of climate risk evaluation and early warning systems.

Altogether, the findings lay a strong analytical ground between atmospheric physics and machine learning performance, which supports the aim of the study to reveal scientifically interpretable and computationally robust predictors of heavy precipitation caused by storms.

7. CONCLUSION

The current research was aimed at exploring the meteorological predictors of storm-related heavy rainage using an organized applied data science framework. The project was able to develop a scientifically interpretable and computationally sound understanding of storm formation dynamics through the integration of real world atmospheric data, statistical analysis, visualization technique, feature selection methods and machine learning modelling.

The main result of the analysis was that the indicators of atmospheric moisture have been identified treatment at specific humidity, dew point, and relative humidity as the most significant predictors of the occurrence of storms. These variables always showed a strong positive correlation with heavy rainfall events of both filter-based and wrapper-based feature selection procedures. Surface pressure, in its turn, had a strong negative correlation that supported the known meteorological principle of storm systems being closely linked to low-pressure zones, which facilitate the accumulation of vertical air and the processes of the condensation of the air.

The modeling step also emphasized the necessity of making suitable choices of assessment measures in the case of unbalanced environmental data. Even the highest overall accuracy of all the classification models was not found to be satisfactory to determine the potential ability in practice. Ensemble-oriented procedures, specifically the Random Forest classifier, were found to perform better in balancing between the storm detection performance and prediction accuracy in terms of high F1-Score and ROC-AUC values. The observation indicates that ensemble learning methods are effective in modelling non-linear interactions in the atmosphere.

In addition to predictive performance, the research was focused on interpretability and correspondence to physical atmosphere processes, such that analytical results were not just statistical contrivances but significant expressions of storm processes. The visualization-based exploratory analysis further enhanced the generation of insights that indicated seasonal patterns, extreme rainfall peaks, and moisture-pressure interactions that are typical of cyclone-prone coastal settings.

Finally, the project shows that the application of data science methodologies can be used to complement traditional meteorological analysis through a quantitative verification of the drivers of storm formation and data predictive frameworks. The knowledge created can be used in better climate risk analysis, preparedness to disasters, and future studies on more advanced modeling tools or imbalance management approaches. On the whole, the research provides a detailed analytical platform on how heavy precipitations can be analyzed and predicted in risky areas.

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Outputs

- 01_data_preparation.ipynb
- 02_exploratory_data_analysis.ipynb.ipynb
- 03_feature_selection.ipynb
- 04_model_training_evaluation.ipynb

Github: meteorological-feature-influence